



The Digital Transition in Algeria: Challenges and Opportunities

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Abstract

The digital transition is a global phenomenon that is redefining the economic, social and political structures of countries. For Algeria, this transformation has become a strategic imperative, not only to modernize its infrastructure and services, but also to diversify its economy and strengthen its competitiveness on international scene. This study aims to analyse in depth the challenges and opportunities associated with the digital transition in Algeria. We look at the challenges inherent in this shift, such as infrastructure, skills, and cybersecurity, while highlighting the many opportunities it presents for economic growth, improving public services, and spurring innovation. The objective is to provide a comprehensive analysis of the current and future dynamics of digitization in Algeria, based on recent data.

Key words: Digital Transition, Challenges, Opportunity.

JEL Classification: O33, O32, O38

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Introduction:

The digital transition is a process of profound transformation and implementation of digital technology of organizations, economies, and societies in order to create new products, services, and processes or to modify existing information by translating business processes into digital form. It is also defined as the integration of digital technologies (artificial intelligence, cloud, big data, Internet of Things) into organizational processes, models and strategies, with the aim of increasing value through innovation, efficiency and user experience, with a focus on efficiency and cost.

The Chartered Institute of Procurement and Supply (CIPS) defines digitization as the redefinition of models, functions, processes and activities by leveraging technological advances to create an efficient digital business environment, an environment in which gains (operational and financial) are maximized and costs and risks minimized (CIPS, 2020). This underscores the need to adopt state-of-the-art tools and technologies, which has become an unavoidable challenge for anyone wishing to remain competitive in the market. Businesses need to adapt to this new model, which means tackling different challenges while seizing the opportunities it offers. Challenges include updating their frameworks and incorporating cutting-edge technologies to maintain competitiveness.

At the same time, this transformation offers the opportunity to improve the customer experience through the personalization of offers and improved



productivity. By adapting to the demands of digitization, companies can strengthen their commitment to their customers and leverage innovative solutions.

Digital transformation in Algeria is a continuous process, characterized by progress and challenges. The country seeks to modernize its economy and management through digital technologies, but its development remains uneven due to structural, regulatory and infrastructural constraints. This research looks at the challenges and opportunities related to the digital transition, while highlighting the barriers to equity and accessibility by formulating the following issue:

To what extent can the digital transformation in Algeria succeed in the face of structural, regulatory, social and cultural challenges, and what strategies must be implemented to accelerate this transformation by involving all economic and social actors?

To answer this problem, the following questions are asked:

- What is the reality of the digital transition in Algeria?
- What are the main weaknesses of the digital transition in Algeria?
- What are the strengths of the digital transition in Algeria?

Objectives of the study: Some objectives can be drawn from this study:

- Analyze the issues, challenges and opportunities of the digital transition in Algeria, in order to propose avenues for improvement to accelerate this transformation.
- To assess the current state of the digital transition in Algeria.
- Propose recommendations for an inclusive and sustainable digital transition
- Help policymakers optimize their digitalization strategies.
- Guide companies and start-ups in the adoption of digital technologies.
- Raise awareness among civil society about the benefits and risks of digital.
- Contribute to academic research on digital transformation in North Africa.

The importance of the study: This study is of strategic importance, both for public authorities and for companies, investors and civil society. The study will help:

- To understand the obstacles and propose concrete solutions.
- To steer public policies towards a global digital transformation.
- Raise awareness among economic actors of the opportunities of digitization.
- Position Algeria as a digital hub in North Africa.

Methodology: The analytical approach used will make it possible to describe and analyze the digital transition in Algeria. The study will be structured as follows

- State of play of the digital transition in Algeria.
- Challenges of the digital transition in Algeria.
- Opportunities of the digital transition in Algeria.

This research is based on a thorough examination and detailed analysis of recent secondary data sourced from national entities (such as reports from the Ministry of Post and Telecommunications, the National Statistics Office, and the Regulatory Authority for Post and Electronic Communications in Algeria) as well as international sources (including publications from the International Telecommunication Union, the World Bank, the OECD, and the UN regarding the



digital transition). The data encompasses statistics related to digital infrastructure, technology adoption, public policies, and socio-economic impacts, thereby facilitating an objective assessment of the challenges and opportunities associated with the digital transition in Algeria. Through a critical and cross-sectional analysis of this data, the research systematically identifies the main challenges (such as infrastructural gaps, regulatory frameworks, or human capital) and opportunities (such as the potential for e-government, knowledge economy, or financial inclusion) that characterize the Algerian digital landscape.

I. Situation of the digital transition in Algeria

The digital transition in Algeria shows a committed but still uneven and has room for improvement dynamic. Since 2020, the country has embarked on a determined acceleration of its digital transformation, driven by strong political will, structural reforms and progressive institutional dynamics.

1. Infrastructure and connectivity:

Updating core ICT infrastructures is the foundation for achieving the goals of digital transformation. This axis encompasses telecommunications networks, data centers as well as the equipment necessary to ensure robust connectivity and equitable access to digital services. It also has the effect of stimulating innovation and fostering the development of digital technologies. Investment in fundamental infrastructure aims to promote the creation of an ecosystem conducive to economic competitiveness and to optimize the benefits of digital for users.

Ensure quality connectivity for all (individuals and households). Goal to achieve is 100% access. This goal is part of the drive to bridge the digital divide, ensuring that all individuals and households can fully benefit from the opportunities offered by digital, while ensuring stable and effective connectivity. According to statistics from the International Telecommunication Union, 24% of the rural population has no access to the Internet, and the Internet penetration rate reaches 72% in the urban area compared to 43% in the rural area (ARPCE, 2024).

The country has significantly expanded its fiber optic network over 200,000 km of fiber optic network, connecting 5.74 million households to the fixed internet, including more than 1.24 million households through fiber-to-the-home (FTTH) technology, and increased its international bandwidth capacity to 7.8 Terabits/second, which facilitates connectivity and deployment of digital services (Haut-Commissariat A La Numerisation, 2024).

1.1.Fiber Optic:

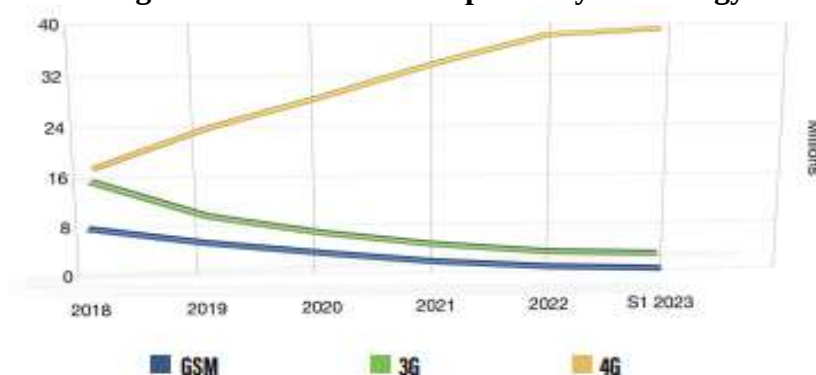
Significant rollout in recent years, gradually covering major cities and some rural areas. The goal is broad national coverage. We are talking about 2.5 million subscribers in 2023, or 18% of households. Goal is to expect 12 million subscribers by 2030 (Haut-Commissariat A La Numerisation, 2024, p. 5).

1.2. 4G/5G: 4G coverage is now very extensive, with around 97% of the population covered by the end of 2023 (ARPCE, 2024). The commercial launch of 5G began in 2024, positioning Algeria among the first in Africa in this area; the



goal is to reach 60% urban coverage by 2027 (Ministère De La Poste Et Des Télécommunications, 2023).

Fig.1. Evolution of subscriptions by technology



Source: (Ministère De La Poste Et Des Télécommunications, 2023)

1.3. Data Centers: Construction and commissioning of several national data centers and cloud services complying with international standards (notably in Algiers, Oran, Constantine) to host data and services locally. The objective is to reach more than five national data centers scattered throughout the national territory, able to meet national requirements (public/private) in terms of processing, storage and hosting. This will help strengthen technological autonomy and digital sovereignty (Haut-Commissariat A La Numerisation, 2024, p. 15).

2. Governmental initiatives

It is a comprehensive roadmap to accelerate the digital transformation of the country (administration, economy, society).

2.1. National Strategy "Digital Algeria":

In 2020, the higher state authorities established an appropriate institutional framework through the creation of the Ministry of Digitization and Statistics (MNS), whose main function is to bring together efforts and integrate the digital transformation process as a strategic axis of the country's new economic development policy

2.2. Electronic Administration (e-gov):

Deployment of platforms and portals such as the Algerian Administration Portal, Single Window of Foreign Trade, online declaration and payment of taxes, request for administrative documents..., and these initiatives aim to eliminate the material aspects of administrative processes. According to the Electronic Administration Index (**e-gov**) for the year 2022, Algeria ranks 126th out of 193 countries, an improvement compared to the year 2020 (141st place) (Nations Unies, 2022, p. 263).

2.3. Key Services (Haut-Commissariat A La Numerisation, 2024):

According to the progress report of the "Digital Algeria 2030" strategy, a document that takes stock of the progress of Algeria's digital transformation. This strategy, launched in 2023, aims to digitize all sectors of the Algerian economy and society by 2030. 67% of administrative procedures have been dematerialized, resulting in



a reduction of processing time from 15 days to just 3 days for the Single Window of Foreign Trade. However, only 28% of the population consistently utilizes online services (Cnis, 2025) .

2.4. Legislative framework:

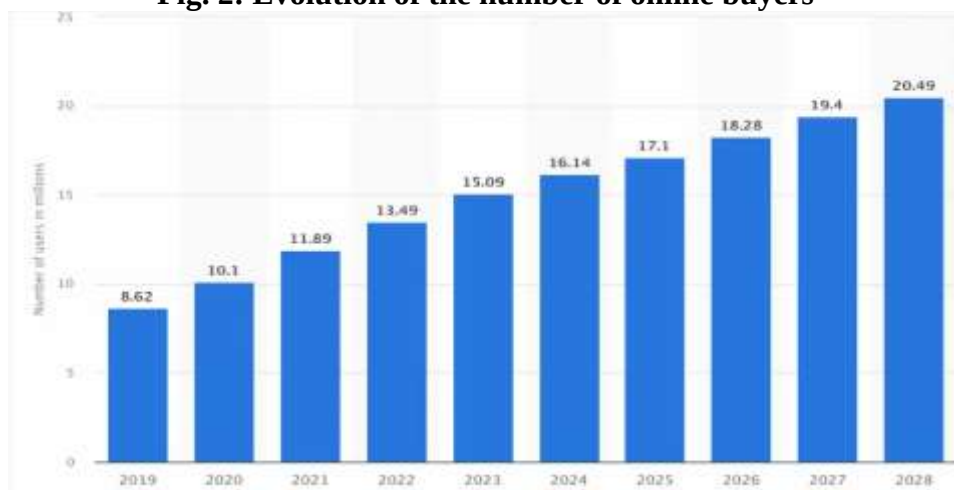
Important laws have been adopted to regulate the digital transition in Algeria, including:

- Law No.18-07 on the protection of personal data. This law, adopted in 2022, led to the establishment of the DPA (Data Protection Authority) in 2023 (joradp, 2018, p. 10).
- Law on the Prevention and Fight against Cybercrime, known as Law No. 09-04 of 5 August 2009, aims to define and punish crimes associated with information and communication technologies (joradp, 2009, p. 04). In addition, it establishes a national body dedicated to the prevention and fight against cybercrime. More than 40% of attacks were recorded between 2021 and 2023 (APS, Adopter une approche rigoureuse pour développer la cybersécurité , 2025) and only 12% of Algerian SMEs have a cybersecurity strategy (APS, La cybersécurité et l'éthique de la profession judiciaire, 2025).
- Law No.15-04 on Electronic Signature and Identification, promulgated on February 1, 2015, establishes the general principles concerning electronic signature and certification in Algeria (joradp, 2015, p. 06).
- Law No.18-05 on electronic transactions which concerns electronic commerce in Algeria and was adopted in 2018, regulates electronic transactions. Its goal is to protect consumers online, ensure data security and foster trust in electronic exchanges (joradp, 2018, p. 04).

3. Digital Economy

Generally speaking, information and communication technologies (ICT) perform two functions in the growth process. First, they participate significantly in capital accumulation. Second, they contribute to the growth of total factor productivity (TFP). This contribution is conditioned by technological advances in the production of goods and services associated with ICT. The information and communication technology sector accounted for 3.2% of Algeria's GDP in 2023, compared to 1.8% in 2015, according to the Ministry of Finance in 2024. This information indicates a significant growth in the share of the ICT sector in the Algerian economy over the period (Banque Mondiale).

The e-commerce sector in Algeria is experiencing remarkable momentum. The number of e-merchants registered at the National Trade Register Centre (NRC) has increased by an average of 92% each year since 2020, reaching nearly 4,000 operators by the end of March 2024. Simultaneously, the quantity of online transactions for products and services experienced a threefold increase from 2020 to the conclusion of 2024, nearing 13 million transactions. In 2023, the value of online sales between businesses and consumers was obtained at an impressive \$1.9 billion, or 262 billion Algerian dinars, despite the fact that 78% of Algerians express distrust of online payments (JUST-INFO, 2025).

**Fig. 2: Evolution of the number of online buyers**

Source: (Statista)

According to the prospective analysis of statista, a well-known statistics site, the population of online shoppers in Algeria is growing steadily, as the following graph reveals. This highlights that the reluctance of Algerians to buy online is gradually decreasing and that this purchasing practice is even beginning to seduce age groups that we never thought would be convinced of this mode of purchase (Statista).

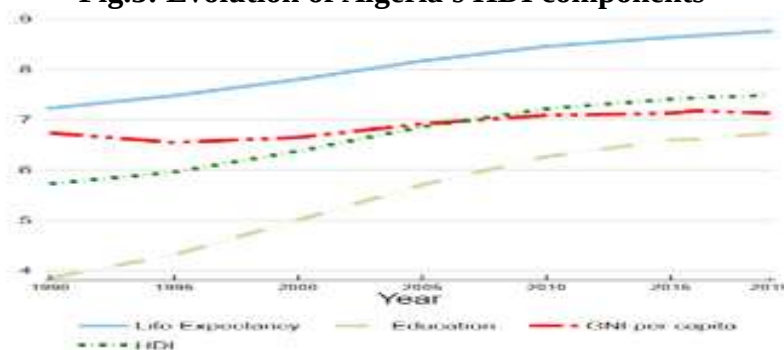
3.1. Startup Ecosystem:

Public policies aimed at anchoring entrepreneurship among young people and in academia, supported by a series of measures promoting the emergence of young energies in the entrepreneurial sector, for his part, the Minister of the Knowledge Economy, Start-ups and Micro-enterprises, Mr. Yacine Mehdi Walid, who participated in a debate session on the challenges of future professions and ways to overcome them, highlighted Algeria's role in Africa in the field of support for startups and innovation. This is manifested in particular through efforts to create an integrated ecosystem for the continent's skills and talents, enacting legislation that encourages investment in knowledge and removing barriers for innovative youth, as established at the first African Startup Conference held in Algeria in 2022 (MESRS, 2023).

4. Human Capital & Skills

Algeria has chosen to enshrine the principles of free education and the democratization of its educational system in all aspects, including the health sector. Significant progress has been made; Algeria is classified in the category of "high human development countries" with a human development index (HDI) of 0.748, ranking 91st in the world among 189 countries and territories, according to the United Nations Development Programme (UNDP) (PNUD, 2020, p. 03).

According to information provided by the Ministry of Higher Education and Scientific Research (MESRS), it is reported that there are 15,000 graduates each year in the information and communication technology (ICT) sector in Algeria, encompassing both universities and schools (MESRS, 2023). However, 45% of companies have difficulty recruiting qualified technical staff.

**Fig.3: Evolution of Algeria's HDI components**

Source: (PNUD, 2020, p. 03)

The Minister of Higher Education and Scientific Research highlighted, in particular, the efforts of the sector which invests in the professions of the future, in particular through the creation of a school dedicated to artificial intelligence as well as the creation of specialized fields in nanotechnology and robotics. The Algerian university is moving towards entrepreneurship and innovation, it should be noted that a third of Algerians continue their studies in schools, universities and vocational training institutes, not to mention the progress made thanks to the free education in the country and the considerable number of schools and universities built.

Algeria's digital transformation is an irreversible strategic choice with solid foundations, but requires better coordination, accelerated implementation and continuity of infrastructure and services to fully exploit the digital potential for the country's economic and social modernization.

II. The challenges of the digital transition in Algeria

The digital transition is a strategic imperative and a key development lever for Algeria, which seeks to become a major player in the digital transformation in Africa. Supported by a strong political will, this initiative is materialized by the National Digital Transformation Strategy (SNTN) to 2030, and aims to modernize the economy, strengthen technological sovereignty and improve the quality of public and private services. Ambitious targets have been set, such as the training of 500,000 ICT specialists and a contribution from the digital sector equivalent to 20% of national GDP (Njaya, 2025). However, this transition is marked by significant challenges. Systemic barriers include:

1. Digital infrastructure:

The country has yet to make sufficient progress in certain areas of fixed and mobile networks. There are gaps in infrastructure and connectivity, including access to reliable and equitable broadband Internet throughout the territory, despite notable advances in infrastructure (fiber optic networks, data centers), bureaucracy, unequal access to the Internet, and delays in training human resources remain major obstacles to homogeneous adoption. Although significant progress has been made in infrastructure (fiber optic networks, data centers), bureaucracy, disparities



in Internet access, and delays in training human resources remain significant barriers to uniform adoption.

2. Cybersecurity:

The term cybersecurity, also known as digital security, refers, in a broad sense, to all the safeguards, measures and technical and non-technical means put in place to protect computer systems and their users against unauthorized access, attacks and damage, in order to guarantee the confidentiality, integrity and availability of data.

According to the Global Cybersecurity Index, Algeria is at level 3 which corresponds to an overall score of at least 55/100 (ITU, 2024, p. 79), the latter shows the fundamental commitment to cybersecurity through actions taken by the Algerian government. These actions include the assessment, establishment or implementation of various generally accepted cybersecurity measures, spread over a moderate number of pillars or indicators.

Cybersecurity is a major concern, given the increase in threats and the need to protect data and information systems. On a small scale, the country is failing to achieve all its goals, although it has begun to develop IT; mentoring IT students and building a legislative framework on cyber security.

3. Digital skills:

The lack of skilled digital skills and resistance to change within organizations are also significant barriers to the adoption and integration of new technologies. Technological evolution is accompanied by a profound evolution of behaviors within structures and processes. There is a form of resistance to change that significantly slows down the integration of technologies in government, business and society. It is important to combat this resistance through awareness-raising and training campaigns.

4. Regulatory and legislative framework:

Current legislation does not always manage to adapt to the speed of technological advances, which hinders innovation and the implementation of digital solutions that meet the requirements of the field. Adjusting the regulatory framework is crucial to establish an environment conducive to innovation and digital trust. The evolution of laws and regulations must take into consideration the digital transformation. This includes data protection and e-commerce legislation, electronic signature as well as intellectual property, to promote innovation and digital trust.

5. Economic and social modernization:

The digital transition is considered a fundamental tool to diversify the economy, encourage innovation and strengthen competitiveness on the international scene. It also aims to reduce territorial and social inequalities by making digital services more accessible. Digital finance, especially for SMEs and startups, often remains inaccessible. It is imperative to develop better mechanisms and promote support in this area.

6. Digital inequality:

There are disparities in access to and use of technologies between the city centre and its periphery, affecting several categories of social classes. To ensure inclusive



change, it is imperative to address this divide to ensure inclusive change (Boustil, 2023, p. 1189).

III. The Opportunities of Digital Transition in Algeria

Despite the challenges, the digital transition offers Algeria unprecedented opportunities for its socio-economic development. The main opportunities arising from the digital transition in Algeria fall into several fundamental areas:

1. Infrastructure Development:

Algeria is dedicated to developing robust foundational digital infrastructures, which include projects like national data centers and a secure sovereign network (APS, La cybersécurité et l'éthique de la profession judiciaire, 2025). This approach effectively addresses the challenges posed by inadequate broadband coverage and antiquated infrastructure by broadening high-speed connectivity and creating resilient, sovereign systems, thus minimizing reliance on foreign networks and improving national cybersecurity.

2. Human capital:

Addressing the challenge of brain drain and graduate unemployment. The goal of training 500,000 ICT specialists is paramount, aimed at limiting the brain drain and providing the country with a skilled workforce to support the digital transformation (Ministère De La Poste Et Des Télécommunications, 2023) serves as a focused response to the dual challenge of youth graduate unemployment and brain drain. By establishing a vibrant digital economy with a genuine demand for these skills, this initiative aims to provide appealing local job opportunities.

3. Digitalization of public services:

More than 500 projects are envisaged for the digitalization of public services, which should improve the quality, transparency and efficiency of interactions between citizens, businesses and administration (Haut-Commissariat A La Numerisation, 2024). By making online procedures traceable and standardized, it diminishes direct interactions that may lead to corruption, enhances transparency, and restores trust between the administration and the citizens, this addresses bureaucratic inefficiency and the absence of transparency by optimizing administrative procedures, reducing paperwork, and facilitating real-time monitoring, thereby enhancing public trust and the quality of service delivery.

4. Emergence of a digital ecosystem:

Algeria aspires to become a regional digital hub, fostering innovation, startups and international partnerships, especially in the field of artificial intelligence by establishing supportive structures, drawing in foreign investment, and positioning Algeria as a competitive entity within the African digital arena.

5. Improvement of business competitiveness

Digitalization offers Algerian companies the opportunity to better manage their data, optimize their internal processes and reduce costs through automation. This translates into improved decision-making, increased productivity, and an ability to respond more quickly to market demands. This reduces low productivity and hesitance towards technology adoption in small and medium-sized enterprises



(SMEs) by offering accessible tools that enhance efficiency, thereby enabling small businesses to compete with larger organizations.

6. Access to new markets, including international

Digital platforms, e-commerce and social networks allow companies to cross local borders to reach a wider customer base, both in Algeria and abroad. This opens up unprecedented prospects for export and commercial expansion. It addresses geographic isolation and restricted export capabilities by facilitating virtual market access, minimizing logistical obstacles, and promoting Algerian products within global value chains.

7. Development of strategic sectors

The digital transition is boosting key sectors such as:

- Transport (example of the start-up Yassir),
- Finance (with the rise of digital banks and the future digital currency of the central bank),
- Health (telemedicine, teleconsultation),
- Agriculture (precision farming, drones).

This tackles the issue of sectoral fragmentation and the lack of innovation within traditional industries by incorporating digital solutions aimed at modernizing operations, enhancing yields, and improving service accessibility in areas that are underserved.

8. Creating and boosting the startup ecosystem

Digital innovation encourages the creation of startups, incubators and technology hubs, contributing to the emergence of a dynamic entrepreneurial ecosystem, generating jobs and local innovation. It addresses the significant issue of youth unemployment and the absence of an entrepreneurial culture by offering platforms that foster innovation, facilitate skill application, and develop scalable business models that are customized to meet local requirements. Beyond 3.3, this specifically addresses the risks of corruption and institutional inertia by digitizing audits and public procurement processes, thereby promoting accountability and drawing foreign direct investment (FDI) through enhanced business facilitation.

9. Modernizing public services and improving governance

The digitalization of administrations simplifies procedures for processes for both citizens and businesses, enhances transparency, and diminishes bureaucratic hurdles, thereby fostering trust and economic appeal. It addresses elevated operational expenses associated with manual tasks by facilitating more efficient operations, thus liberating resources for reinvestment in areas of growth.

10. Reduced transaction costs & resource optimization.

Automating tasks decreases manual labor reduces errors and costs associated with physical data storage, while increasing efficiency.

11. Flexibility and agility in the face of market developments

Digital technologies offer companies the opportunity to adopt agile methods, quickly adapt to transformations and continuously innovate, which is a significant competitive advantage. This mitigates susceptibility to external disturbances and sluggish market reactions by providing enterprises and organizations with



immediate data and flexible systems, thereby guaranteeing robustness in an unpredictable global economy.

The digital transition in Algeria represents a source of considerable opportunities, ranging from enhanced competitiveness, access to new markets, modernization of key sectors and the creation of an innovative ecosystem. Properly exploiting these opportunities can foster sustainable and inclusive economic growth.

Conclusion

The digital transition in Algeria is part of a global dynamic of transformation of societies and economies. This process, although uneven across sectors, represents a key strategic issue for the country's development.

Algeria is embarking on an ambitious and structured digital transition, with significant challenges regarding economic, social and institutional modernization, as well as considerable opportunities related to infrastructure development, training and the digitization of public services. The Algerian government has implemented a strategy based on five fundamental axes (Haut-Commissariat A La Numerisation, 2024):

- Development of basic infrastructure: This involves expanding the fiber optic network, improving internet connectivity and deploying 5G to ensure broadband access throughout the territory.
- Human Resources and Training: The focus is on building digital skills at all levels, from primary education and vocational training to specialization in advanced areas such as artificial intelligence and cybersecurity.
- Modernization of public administration and services: The objective is to dematerialize administrative procedures, develop e-government, e-health and e-education in order to simplify the lives of citizens and businesses.
- Development of the digital economy: This axis aims to promote the creation of innovative startups and SMEs, to support investment in the information and communication technologies (ICT) sector, and to encourage exports of digital services.
- Cybersecurity and data protection: The establishment of a solid legal and technical framework to ensure the security of information systems and the protection of personal data is essential.

The bureaucracy within Algerian administration, marked by slowness and a lack of coordination, a digital culture still underdeveloped, as well as a rigid regulatory framework, represents a significant obstacle to the digital transition.

In order to accelerate the implementation of initiatives, it is essential to simplify procedures, intensify training, improve governance and adjust regulations to digital realities. The transition must serve as a catalyst to modernize public governance, improving the transparency, accountability and efficiency of administrations, while laying the foundations for national digital sovereignty.

From this brief analysis of the current state of the digital transition, we conclude that the expansion of the technology sector and digitization rest on four essential



foundations: access to broadband connections, the facilitation of trade (whether domestic or international), and the interaction and perception of digital technology by the state, businesses and citizens. It is important to note that access to broadband connections is the key to ensuring all other activities necessary for a smooth and successful digital transition. The success of this process will depend on the ability to overcome barriers related to infrastructure, governance and digital inclusion, in order to position digital as a driver of sustainable growth and innovation.

Recommendations

We make several recommendations to support the strengthening of the digital transition in Algeria:

- Investing in infrastructure and social assistance;
- Develop training programs for human resources;
- Strengthen applied research and innovation capacities;
- Promote international academic exchanges;
- Improve coordination and evaluation of the digital transition;
- Integrate artificial intelligence (AI) to develop local solutions and reduce technological dependence.

These transformations can help make Algerian industry a more efficient, agile and integrated sector in the global digital economy.

Bibliographic References :

- APS. (2025, 06 25). Consulté le 07 11, 2025, sur La cyber sécurité et l'éthique de la profession judiciaire: <https://www.aps.dz/algerie/tag/Cybers%C3%A9curit%C3%A9>
- APS. (2025, 07 05). *Adopter une approche rigoureuse pour développer la cyber sécurité*. Consulté le 07 11, 2025, sur APS: <https://www.aps.dz/algerie/tag/Cybercriminalite>
- ARPCE. (2024). Consulté le 07 08, 2025, sur <https://www.arpce.dz/fr/indic/mobile>
- Banque Mondiale. (s.d.). Fondations pour le developpement des technologies de l'information et de la communication en algerie. Consulté le 07 11, 2025, sur: <https://documents1.worldbank.org/curated/en/696081468767408975/txt/258410FRENCH0Algerie0Note0TIC.txt>
- Boustil, Z. (2023). La fracture numérique en Algérie : Etat des lieux et perspectives. *Revue recherches et études en sciences humaines* , p 1182-1201.
- CIPS . (2020). Consulté le 07 05, 2025, sur Numérisation des achats et de l'approvisionnement 2020: <https://www.cips.org/intelligence-hub/member-only/procurement-technology-digitalisation-guide>
- Cnis. (2025, 02). *Bilan du Moyen terme 2019-2023*. Consulté le 07 08, 2025, sur Rapport: <https://www.cnis.fr/wp-content/uploads/2025/03/rapport-169-bilan-du-moyen-terme-2019-2023.pdf>
- Haut-Commissariat A La Numerisation. (2024, 08). *Stratégie Nationale de la Transformation Numérique*. Consulté le 07 07, 2025, sur <https://hcn.dz/SNTN->



Fr.pdf?fbclid=IwY2xjawKOqfBleHRuA2FlbQIxMABicmlkETFKZHFndmlBc3FGRDRVRlpHAR6cvX_520xCPVunds3nKZYuQaO-d4qWhph9j2zfwPMPJ45THsrBoeu19HT5Sg_aem_1fTOugsHA9trVqtXYjQe9A

- ITU. (2024). *Global Cybersecurity Index 2024*. Consulté le 07 14, 2025, sur ITU Publications: file:///C:/Users/win7/Downloads/d-hdb-gci.01-2024-pdf-e.pdf
- Joradp. (2009, 08 05). Consulté le 07 25, 2025, sur <https://www.joradp.dz/ftp/Jo-francais/2009/f2009047.pdf>
- Joradp. (2015, 02 10). Consulté le 07 25, 2025, sur <https://www.joradp.dz/FTP/jo-francais/2015/F2015006.pdf>
- Joradp. (2018, 06 10). Consulté le 07 25, 2025, sur <https://www.joradp.dz/FTP/JO-FRANCAIS/2018/F2018034.pdf>
- joradp. (2018, 05 16). Consulté le 07 25, 2025, sur <https://www.joradp.dz/FTP/jo-francais/2018/F2018028.pdf>
- JUST-INFO. (2025, 06 25). *JUST-INFODZ*. Consulté le 07 11, 2025, sur Algérie : le e-commerce décolle... sans pilote: <https://just-infodz.com/algerie-le-e-commerce-decolle-sans-pilote-rapport/#:~:text=Les%20chiffres%20parlent%20d'eux%2Dm%C3%AAs%20:%20le%20commerce,de%204%20000%20op%C3%A9rateurs%20fin%20mars%202024>.
- MESRS. (2023). Consulté le 07 13, 2025, sur <https://www.mesrs.dz/index.php/agregats-ar-2024-2025-2/>
- Ministère De La Poste Et Des Télécommunications. (2023). *RAPPORT DES INDICATEURS TIC*. Consulté le 07 08, 2025, sur <https://www.mpt.gov.dz/chiffres-et-indicateurs/rapport-des-indicateurs-tic/>
- NATIONS UNIES. (2022). *Enquête sur L'E-gouvernement 2022*. Consulté le 07 08, 2025, sur <https://publicadministration.un.org/egovkb/Portals/egovkb/E-Government%202022%20Sep%2027%20Fr%20Final%20-DK.pdf>
- Njaya, S. (2025, 05 13). *L'Algérie dévoile sa stratégie de transformation numérique à l'horizon 2030*. Consulté le 07 13, 2025, sur <https://www.wearetech.africa/fr/fils/actualites/tech/lalgerie-devoile-sa-strategie-de-transformation-numerique-a-lhorizon-2030>
- PNUD. (2020). *Rapport sur le développement humain 2020*. Consulté le 07 13, 2025, sur <https://hdr.undp.org/sites/default/files/Country-Profiles/fr/DZA.pdf>
- Statista. (s.d.). *Etudes et rapports*. Consulté le 07 01, 2025, sur <https://fr.statista.com/etudes-et-rapports/>
- Tech Hub. (2025, 03). *Impact de la digitalisation sur les entreprises en Algérie*. Consulté le 07 13, 2025, sur <https://ikosim-tech-hub.com/impact-de-la-digitalisation-sur-les-entreprises-en-algerie/>